Bonus Project Deep Learning (SS26): Multivariate Time Series Forecasting Exposé

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1 Idea

This project investigates deep learning approaches for multivariate time series forecasting and evaluates them on the provided benchmark dataset. We compare recurrent-attention hybrid structures against purely attention-based transformer architectures, analyzing their ability to capture both localized short-term dynamics and long-range temporal dependencies across correlated variables.

2 Planned Architectures

We consider two main model families for implementation, optimization, and evaluation:

1. **LSTM + Attention (Recurrent-Attention Hybrid):** The first approach utilizes an encoder-decoder architecture based on Long Short-Term Memory (LSTM) networks [1] combined with a temporal attention mechanism [2]. The motivation behind this choice stems from the inherent strength of LSTMs in capturing sequential temporal dynamics across hidden states, while the integrated attention modules help mitigate the classic vanishing gradient issues of long sequences by directly focusing on the most critical historical timesteps.
2. **Patch-Based Transformer (PatchTST):** The second approach uses *PatchTST* [3], a modern Transformer optimized for time series that avoids the high cost and point-wise semantic loss of vanilla Transformers [4]. It splits each series into sub-series patches, retaining local semantics and reducing attention complexity from $O(L^2)$ to $O((L/P)^2)$ for patch length P. It also employs channel-independence: each variate is processed as a separate univariate sequence by a shared Transformer backbone, a design shown to improve generalization on multivariate benchmarks [3].

3 Additional Dataset

To assess generalization beyond the benchmark, we use the **Jena Climate Dataset** [5], compiled by the Max Planck Institute for Biogeochemistry, which records 14 atmospheric features (air temperature, pressure, humidity, wind direction, etc.) at 10-minute intervals. We frame the task as rolling-horizon forecasting of future temperature and pressure profiles from historical climate vectors. The non-linear cross-variable dependencies in weather systems make this a rigorous testbed for evaluating PatchTST's [3] multi-channel handling against the LSTM-Attention baseline.

4 Planned Improvements

Planned ablations and optimizations include:

- **Training dynamics:** Scheduled sampling [6] for the LSTM decoder to interpolate between teacher forcing and free-running decoding, coupled with Cosine Annealing with warm restarts (SGDR) [7] for convergence stability.
- **Normalization and regularization:** RevIN (Reversible Instance Normalization) [8] to handle distributional shift, with Spatial Dropout [9] against overfitting.
- **Ensembling:** A weighted blending ensemble combining the LSTM-Attention's stability with PatchTST's [3] capacity.

Models will be compared against the course baselines (Naive last-value, Lag-24/Lag-168, Seasonal mean) via MAE and MSE.

5 Risks and Open Questions

- **Error propagation:** Long autoregressive rollouts risk error accumulation; we will test whether PatchTST's direct multi-step output mitigates this versus iterative LSTM decoding.
- **Noise vs. signal:** High-frequency noise can mislead global self-attention; choosing a patch length that filters local noise while preserving global structure is an open hyperparameter question.
- **Efficiency–accuracy trade-off:** Whether the cost of a fully attention-based network yields gains that beat a well-tuned LSTM-Attention hybrid on the private test set.

References

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